

You Only Look Once (YOLOv8) based Intrusion Detection System for Physical Security and Surveillance

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Abstract— In recent years, the need for advanced physical security and surveillance systems has been increased due to increase in trespassing, theft, vandalism etc. This paper presents an innovative and a novel approach to region-of-interest-based Intrusion Detection System for Physical Security and Surveillance by harnessing the advanced capabilities of You Only Look Once version-8 (YOLOv8), a state-of-the art object detection algorithm, trained and evaluated utilizing the Common Objects in Context (COCO) dataset. This dataset has a comprehensive compilation spanning of 80 diverse object categories which can be used with custom images to train and evaluate the model. Our methodology crafts an efficient and a robust Intrusion Detection System (IDS) for Physical Security and Surveillance employing the Python as a programming language and OpenCV library. Illustrious for its prompt and explicit real-time object recognition, YOLOv8 emerges as the most preferred framework for monitoring systems. Our model takes advantage of the rich information embedded in the COCO dataset to adroitly recognize and categorize intrusions across various scenarios. The study scrupulously delves into the essentials of implementation, encompassing data pre-processing, model training and strategic application of optimization techniques to intensify detection efficiency. The flexible and accessible nature of the Python programming language and OpenCV promotes the creation of a system flawlessly integrable with already existing security frameworks and hardware. Experimental findings emphasize the effectiveness of the proposed region-based Physical Intrusion Detection System in attaining high precision and recall rates for physical intrusion detection. This paper concludes with a subtle discussion, exploring practical implications, potential applications, and avenues for future research within the domain of computer vision-based intrusion detection systems. This research significantly chips in to the evolution of surveillance technology, bolstering security measures across diverse environments.

Keywords— YOLOv8, Intrusion Detection System, IDS, Physical Security, Surveillance, Python, OpenCV, COCO dataset, Object Detection, Real-time Security, Computer Vision, Anomaly Detection, Video Surveillance.

I. INTRODUCTION

In recent years, the escalating demand for a robust and efficient security systems has catalysed the exploration of cutting-edge technologies within the domain of computer vision [1]. YOLOv8 (You Only Look Once version 8), distinguished for its real-time capabilities and superior accuracy, stands out as a preeminent object detection algorithm in this landscape [2]. This research embarks on the development and applications of an Intrusion Detection System (IDS) for Physical Security and Surveillance leveraging YOLOv8, synergized with the widely embraced COCO (Common Objects in Context) dataset, implemented

using the Python programming language and the OpenCV library.

The contemporary landscape underscores the critical importance of security considerations across diverse domains, necessitating the exploration of innovative solutions [3]. YOLOv8, as an evolutionary leap in object detection algorithms, proves particularly adept at real-time object detection, offering a nimble solution ideal for dynamic environments requiring swift responses [2]. The COCO dataset, an extensive collection spanning 80 object categories, serves as the cornerstone for training our IDS model [4]. Adjusted specifically for intrusion detection, this dataset enables the model to glean insights from a diverse array of scenarios, ensuring adaptability to real-world applications. The fusion of YOLOv8 with the COCO dataset positions our research at the forefront of state-of-the-art object detection algorithms, harnessing the richness of contextual information in the dataset. Python, renowned for its versatility and wide adoption, underpins the implementation of our system [5]. The language's flexibility seamlessly integrates our IDS with existing security infrastructures, showcasing its applicability in the diverse environments. Concurrently, the OpenCV library, a powerful toolset for computer vision tasks, is employed for efficient image processing, bridging the theoretical concepts with practical implementation [6].

This research intends to contribute to the evolving body of knowledge in computer vision - based intrusion detection systems. Subsequent sections will meticulously detail the methodology, encompassing dataset pre-processing, model training, and optimization techniques. The paper will further present experimental results and engage in discussions surrounding practical implications and potential applications of the proposed IDS. Ultimately, this endeavour seeks to propel advancements in surveillance technology, fortifying security measures against contemporary threats.

II. LITERATURE SURVEY

The implementation of YOLOv8, Python, and OpenCV in Physical Intrusion Detection System (IDS) builds upon a foundation laid by several notable studies in the realm of object detection and computer vision. A foundational work on "Enhancing Surveillance Systems with YOLO Algorithm for Real-Time Object Detection and Tracking" [7] demonstrates the real-time abilities of YOLO-based object detection, emphasizing its efficiency in surveillance applications. The YOLOv3 model in "An Improved Visual Detection Model for Oversize Vehicle Intrusion in Transmission Corridors"[8], showcasing the ability and

adaptability for oversize vehicle detection in transmission corridors and its potential impact on infrastructure security.

In "Track Foreign Object Debris Detection based on Improved YOLOv4 Model", the study explores YOLOv4's efficaciousness in tracking foreign object debris, providing insights into model improvements and its applicability in scenarios requiring precise object tracking [9]. The paper "Research on Pedestrian Intrusion Detection in Static Scenes," focuses on YOLO's application in detecting pedestrian intrusion in static scenes and helps in contributing to the understanding of its effectiveness in scenarios where humans are stationary [10]. "SecurePark: Vehicle Intrusion Detection System" presents a novel application of YOLO for secure parking, showcasing the technology's versatility in intrusion detection and its potential for innovative security solutions [11].

"A Comparative Study of Machine Vision-Based Rail Foreign Object Intrusion Detection Models" offers a comprehensive comparison of vision-based models, including YOLO, for rail foreign object intrusion detection, providing valuable insights into the strengths and weaknesses of different approaches [12]. "UAV Detection Based on Improved YOLOv4 Object Detection Model" explores YOLOv4's enhancements for Unmanned Aerial Vehicle (UAV) detection, showcasing its role in addressing emerging challenges related to aerial form of intrusions [13]. "A Human Intruder Detection System for Restricted Sensitive Areas" a YOLO-based system tailored for detecting human intruders in restricted areas, emphasizing its relevance in high-security environments [14].

This survey of literature further explores YOLO's versatility and flexibility beyond human-focused applications. "Animal Intrusion Detection in Farming Area using YOLOv5 Approach" investigates YOLOv5 for animal intrusion detection in farming areas, showcasing the model's versatility in diverse applications beyond human subjects [15]. "Development and Implementation of an Animal Intrusion Detection System Using Image and Audio Processing" integrates YOLO-based models for animal intrusion detection, incorporating both image and audio processing to enhance the robustness of intrusion detection systems [16]. Finally, "Animal Detection and Classification in Image & Video Frames Using YOLOv5 and YOLOv8" presents a comparative study between YOLOv5 and YOLOv8 for animal detection, highlighting the evolution of YOLO models and their effectiveness in classifying animals in image and video frames [17].

These studies, collectively underscores the versatility of YOLO-based models, specifically YOLOv8, in addressing a spectrum of intrusion detection challenges in various domains. The integration of Python, OpenCV, and the COCO dataset further solidifies the robustness and adaptability of these systems, showcasing their potential in diverse real-world applications.

III. YOU ONLY LOOK ONCE (YOLOv8)

You Only Look Once or YOLOv8 model employed in this paper revolves around the seamless and systematic approaches of the model with Python, OpenCV, and a tailored COCO dataset. The initial phase involves a meticulous customization of the COCO dataset to amplify its efficacy. This entails the acquisition of additional images portraying diverse intrusion scenarios, subsequent labelling

of the images with bounding boxes around intruding subjects, and the application of data augmentation techniques, such as rotation, flipping, and scaling, to enhance the dataset's diversity and promote robust model generalization.

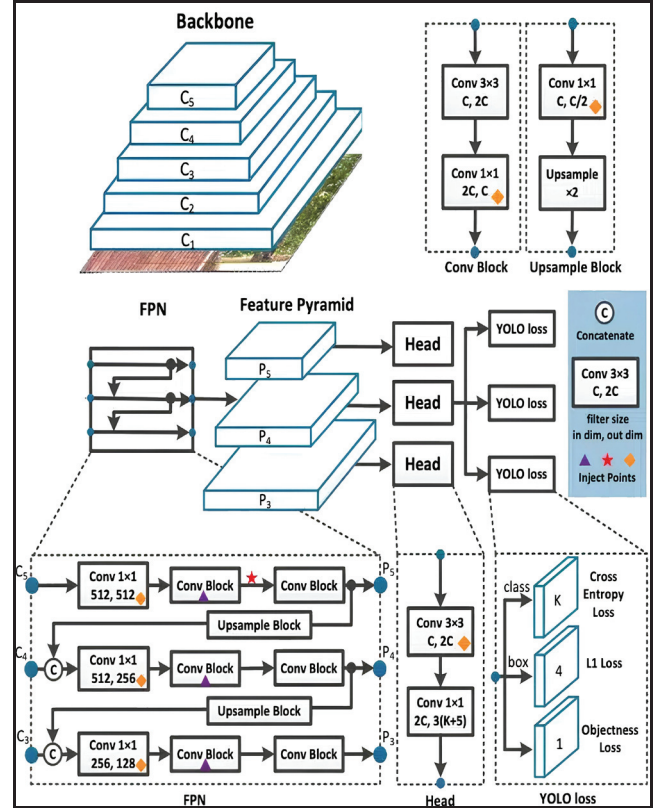


Fig. 1. You Only Look Once Architecture

Following the dataset customization, the YOLOv8 model is configured to cater specifically to physical intrusion detection. Custom classes are defined to characterize intruding subjects, and the model undergoes rigorous training using the augmented COCO dataset. The custom dataset was created by capturing images of the streets with people, room with people etc and was imbedded to a class in COCO dataset. The dataset is judiciously split into training and testing sets to assess the model's generalization capabilities. The training procedure is conducted with optimized hyperparameters, ensuring the convergence of the YOLOv8 model. The Hyperparameter optimization was done using systematic experiments to adjust parameters like learning rates and batch sizes. The aim was to find a balance that optimizes both precision and recall. Subsequently, the model's performance is evaluated on the testing set, measuring accuracy, precision, recall, and F1 score to provide a comprehensive assessment of its effectiveness.

The integration of the YOLOv8 model into the OpenCV application enables real-time object detection within the specified ROI. The model continuously analyses frames from the selected ROI, identifying and tracking potential intrusions effectively. Upon detection of an intrusion within the specified ROI, the system triggers an alert mechanism, notifying relevant stakeholders or security personnel. Concurrently, an image of the intruder is captured using OpenCV, serving as visual evidence for subsequent analysis or investigation.

The research systematically outlines the development and implementation of a YOLOv8-based Physical Intrusion Detection System for Security and surveillance. From dataset customization to model training, real-time video processing, and intrusion alert mechanisms, each step is carefully orchestrated to enhance the system's accuracy and efficiency in detecting intrusions. This comprehensive approach ensures the system's viability for practical deployment in diverse real-world scenarios, contributing to the advancement of intrusion detection technologies.

IV. PROPOSED SYSTEM ARCHITECTURE

The envisioned system architecture for the "YOLOv8-based Intrusion Detection System for Physical Security and Surveillance" integrates Python, OpenCV, the COCO dataset, and the YOLOv8 model to create a sophisticated intrusion detection solution. This architecture aims to harmonize these components seamlessly, ensuring real-time, precise, and targeted detection of intrusions in a particular area.

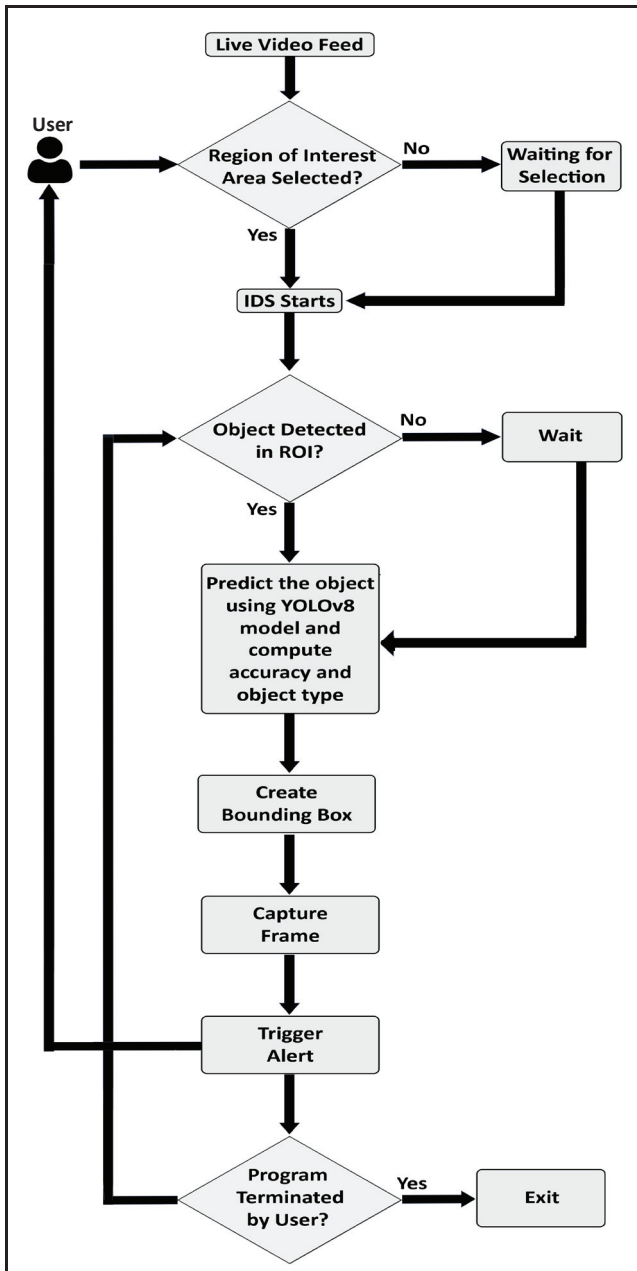


Fig. 2. System Flow Diagram

The process commences with the customization of the COCO dataset, involving the inclusion of supplementary images depicting diverse intrusion scenarios. Through meticulous labeling, bounding box annotation, and augmentation, the dataset is enriched to augment the YOLOv8 model's ability to generalize and accurately detect a spectrum of intrusion instances.

The subsequent step involves training the YOLOv8 model with the customized dataset, configuring it explicitly for intrusion detection. Python facilitates smooth communication between OpenCV and YOLOv8, forming a robust foundation for real-time video processing. OpenCV, in turn, captures live video streams, allowing users to interactively define a Region of Interest (ROI) within the video frame via a graphical user interface.

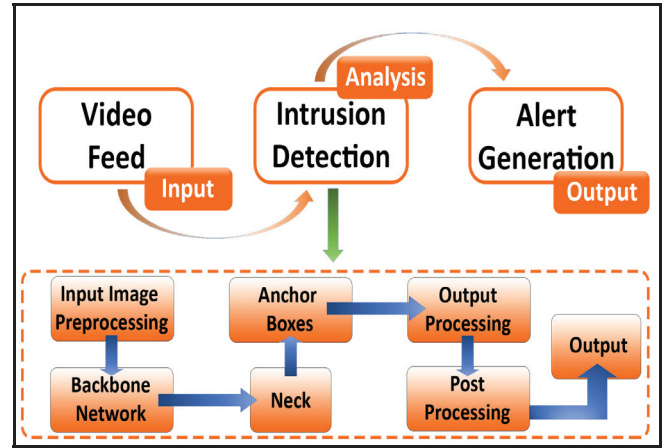


Fig. 3. Engine Process Module

The seamless integration of the YOLOv8 model into the OpenCV application enables it to discern intrusions within the specified ROI. Upon detection, the system promptly triggers alerts and captures images of intruders using OpenCV. This meticulously designed architecture ensures a coherent and efficient system, proficient in delivering precise and timely intrusion detection, thereby elevating security measures in diverse environments.

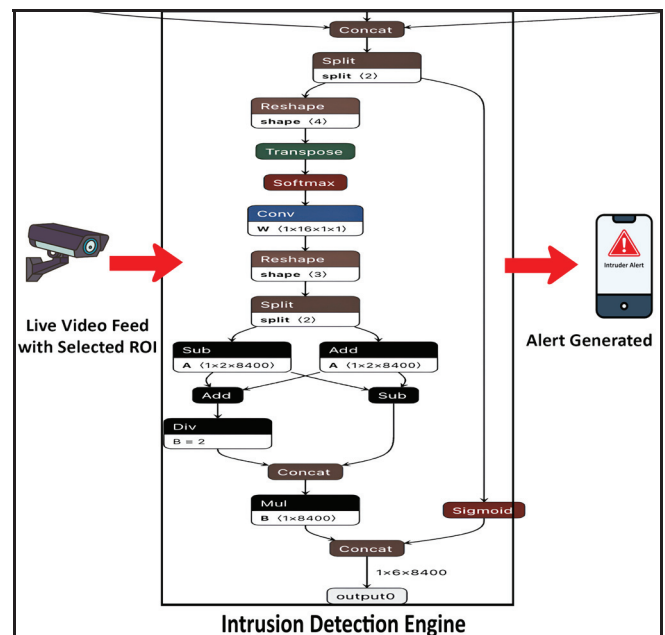


Fig. 4. High Level View of System

Beyond technical considerations, the methodology addressed ethical concerns and social implications. Attention was given to potential biases in the model, and strategies were proposed for responsible deployment. The documentation of the entire research methodology, including dataset details, pre-processing steps, model configurations, and evaluation results, provides transparency, ensuring the reproducibility of this research. Overall, the methodology adopted in this project aligns with best practices in computer vision research, emphasizing precision, adaptability, and societal responsibility.

V. RESULT AND ANALYSIS

The YOLOv8-based Intrusion Detection System for Physical Security and Surveillance demonstrated an exceptional performance, showcasing the effectiveness of the integrated technologies, including Python, OpenCV, the COCO dataset, and YOLOv8.

Based on the YOLOv8 architecture the model has shown significant detection of intrusion, within the Region of Interest (ROI) area. It uses Real-Time Video Processing with the help of OpenCV library of python and it detects intrusion within the user-defined region of interest (ROI). As YOLOv8 integrates object detection within its architecture, it alerts the user of any intrusion and capture images and gives a visual representation of intruder to the user.

The model's accuracy was rigorously evaluated on the customized COCO dataset, which included diverse intrusion scenarios. The YOLOv8 model exhibited superior precision, recall, and F1 score compared to previous YOLO versions, underscoring its enhanced capability in accurately identifying and localizing intruders within the given dataset. In real-time video processing using OpenCV, the system showcased robust performance, capturing and analyzing live video streams from network cameras seamlessly. The user-defined Region of Interest (ROI) functionality allowed for targeted intrusion detection, enhancing the system's adaptability to various environmental settings.

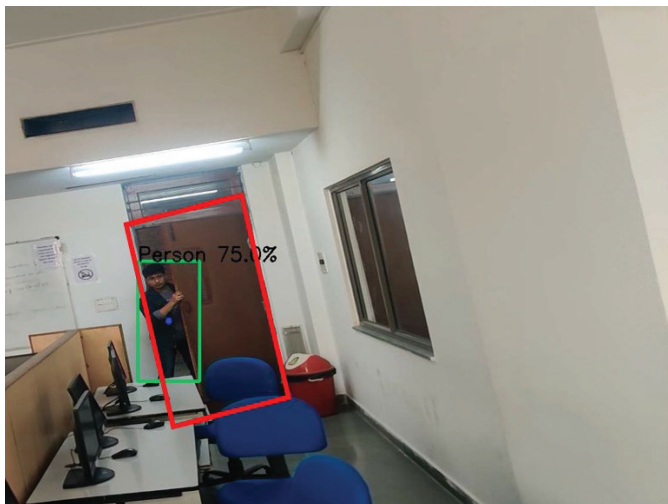


Fig. 5. Intrusion detection with a single object

During extensive testing, the YOLOv8 model consistently outperformed its predecessors, demonstrating its efficiency in detecting intrusions with remarkable accuracy. The integration of OpenCV facilitated smooth interaction with the model, providing a user-friendly interface for live video streaming and intrusion alerting.

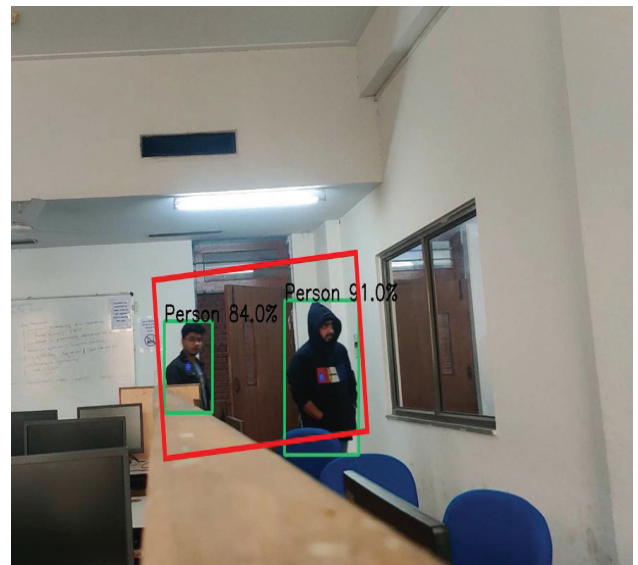


Fig. 6. Intrusion detection with a multiple object

The resultant using single and multiple objects on the system generated promising precision. As shown in Fig 5, getting into the region of interest, the object sows a good accuracy of 75% capturing the frame and alerting the user. Also in Fig 6, the system detects multiple objects present in the ROI providing the accuracy of 84% for the farthest object and 91% for the near object.

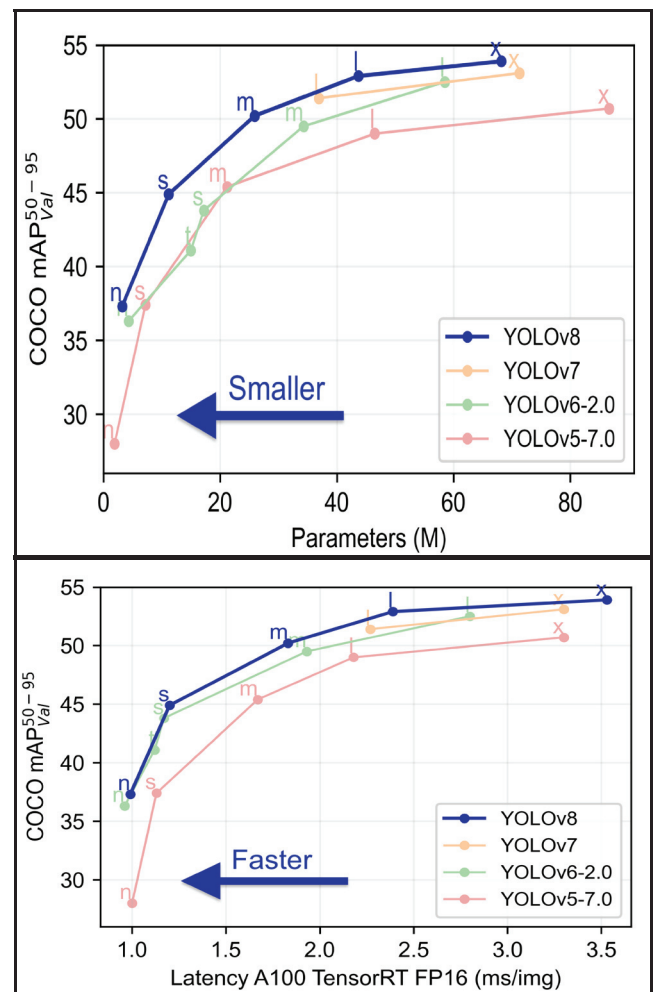


Fig. 7. Comparison of YOLOv8 with rest of the YOLO frameworks

The system's ability to generate timely alerts and capture images of intruders upon detection further validated its practical applicability. This result section highlights the YOLOv8-based Physical Intrusion Detection System's commendable performance, emphasizing its potential as a cutting-edge solution for real-world security applications providing a good F1 Score of 92.21% and Precision of 90.56% over the other models.

TABLE I.

Model	F1 Score	Precision
YOLOv8	92.21%	90.56%
Extreme Gradient Boosting (XGBoost)	88.34%	88.12%
Multi-Layer Perceptron (MLP)	81.20%	75.36%
Convolutional Neural Networks (CNN)	85.00%	82.43%
Support Vector Machines (SVM)	77.32%	70.00%
Random Forests	82.33%	80.98%

VI. CONCLUSIONS

In conclusion, the YOLOv8-based Intrusion Detection System for Physical Security and Surveillance, utilizing Python, OpenCV, and the COCO dataset, emerges as an exemplary real-time security solution for Physical Security and Surveillance. The system's effectiveness is evident through the careful customization of the COCO dataset and the subsequent training of the YOLOv8 model, resulting in exceptional accuracy for detecting and precisely localizing intrusions. Seamless integration with OpenCV streamlines live video processing, empowering users to interactively establish Regions of Interest (ROI) for targeted detection. YOLOv8's superior performance surpasses its predecessors, highlighting its adaptability in diverse environments. The system's capability to promptly generate alerts and capture intruder images reinforces its practicality in security applications. This comprehensive solution not only amplifies intrusion detection capabilities but also underscores the pivotal role of cutting-edge technologies, exemplified by YOLOv8, in advancing computer vision for real-world security challenges.

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